Saumya Mehta

AI Engineer

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Senior AI Engineer with 6+ years designing and shipping production LLM, RAG, and recommendation systems across edtech, banking, and healthcare. Presented at ICICLE-25 on large-scale experimentation → 1.2 × faster iteration for educators. Proven ability to operationalise AI solutions, communicate complex technical ideas to non-experts, and lead cross-functional initiatives.

WORK EXPERIENCE

Randstad Digital, Client: Bank Of America — Sept 2025 - Present
Data Scientist, Machine Learning & AI — Plano, TX

- Architected a **multi-task learning framework** to classify 21 document types across 4 sources, using **PCgrad** and **homoskedastic uncertainty weighting** to resolve conflicting optimization paths. This raised straight-through processing to **49%** (at 95% confidence) and increased document accuracy from 67% to 76%.
- Developed a multi-agent system using **group deliberation methods** for LLM trustworthiness, employing an ensemble of models (**LLaMA 3.1 70B**, **GPT-OSS**) to calibrate confidence. Achieved significant performance gains: **35% reduction in Expected Calibration Error (ECE)** and **20% improvement in the Brier score**.

Aseesa Inc — Sept 2025 - Present
AI Research Engineer — Plano, TX

- **Architected** a scalable RAG pipeline to construct a comprehensive **disease knowledge base** from over **5k** scientific articles sourced from **PubMed** and **Crossref**, establishing a high-fidelity corpus for downstream biomedical research.
- **Built and tuned a low-latency vector search stack on Qdrant**, indexing over 10k documents with 50-dim embeddings and rich payload filters, sustaining **5,000+ vectors per second** during ingestion and delivering p95 query latency under 40 ms while improving **precision@10 by 20%** over a BM25 baseline.

Carnegie Learning — June 2023 – July 2025
Senior Data Scientist, Machine Learning & AI — Pittsburgh, PA

- Led RCTs and A/B tests across **10K+ students** in partner school districts, delivering insights that drove core MATHstream feature improvements and informed district-level product decisions.
- Designed and deployed scalable logging infrastructure to capture and standardize **500K+ student events daily** in real time, enabling robust analytics and cross-team research integration.
- Conducted **quasi-experimental analysis** to assess **MATHstream's** impact on standardized test scores, revealing **4–7% gains** in standardised test outcomes resulting in **15% upsell** conversion in pilot districts (est **1.7B revenue**)
- Productionized **real time recommendation algorithms as** Python FastAPI microservices with Pydantic validation and pytest suites; reducing **p95 latency by 57%** using batching, caching, and lightweight quantization
- **Built scalable vector indexing and semantic search systems** leveraging **Pinecone** and OpenAI embeddings, optimizing document chunking, similarity retrieval, and response latency for downstream LLM tasks.
- Built Python NLP pipelines for NER, document classification, and summarization to index curriculum and parse student responses, powering recommendations and adaptive hint generation; **HR@5 +12%, NDCG@10 +9%**
- Built PySpark + Airflow pipelines for recency/mastery/difficulty features; improved data freshness from hours to minutes, lifting rec quality on new content.
- **Designed and deployed Retrieval-Augmented Generation (RAG) pipelines** using LangChain, OpenAI embeddings, and PGVector to enable contextual response generation from proprietary document datasets.

Pearson(Savvas) — May 2022 – Aug 2022
Software Development Engineer Intern — Boston, MA

- Developed **Curriculum Recommendation** algorithms for the SuccessMaker engine using **Bayesian Item Response Theory** resulting in a **40%** improvement in student test scores
- Streamlined the data analytics pipeline for the Learning Analytics team by creating external views in **Amazon Redshift DB** and **automated query scheduling** on AWS for ETL tasks

Playpower Labs — Jun 2018 – Aug 2021
Data Scientist — Remote

- Led the research and development of an **AI-powered** paper-based formative learning product aimed at creating equitable assessments for students across India and piloted across **50+** schools
- Built an OCR/Hand written text extraction pipeline with **de-skew/denoise** and **region segmentation** to extract free text responses from paper assessments; achieved **3–5% CER** on clean scans and **6–8% CER** on classroom scans using lexicon-based post-correction.
- Architected and deployed end-to-end machine learning pipelines on **Snowflake**, handling up to **1 million records per second** and delivering a **25%** improvement in model performance and a **50%** reduction in data processing time
- Enhanced real-time data processing capabilities by utilising **Spark Streaming**, **R**, and **Kafka**, delivering a **50%** improvement in processing speed and scalability for a distributed computing setup handling over **100K** records per second of streaming data
- Implemented and optimized **ETL** pipelines using **Apache Airflow** and **AWS Glue**, delivering a **30%** reduction in ETL runtime and improving data quality and reliability
- Implemented custom OpenCV kernels with native functions to deliver sub-second page detection, region extraction, and answer extraction for both free-text and MCQ responses, processing **10–20 page workbooks in under 30 seconds** with around **98% end-to-end grading accuracy**
- Researched and applied model quantization techniques for character-recognition models, reducing model footprint enough to run reliably on **ultra–low-end Android phones in India (sub $50 devices)**.

PUBLICATIONS

Gastroenterology in the age of AI: Bridging technology and clinical practice [doi: 10.3748/wjg.v31.i36.110549] — Sept 2025
World Journal of Gastroenterology

MATHstream and UpGrade: Using Rapid, Large-Scale Experimentation for Data-Driven Improvements [Paper] — June 2025

International Consortium for Innovation and Collaboration in Learning Engineering (ICICLE)

Simvastatin + Metformin Treatment Targets Growth and Fibroinflammatory Responses in PaSc [Poster] Dec 2023
American Pancreatic Association

Advanced Chemical Transport Modeling in Dynamic Multicellular Contexts Using CompuCell3D [Manuscript] May 2023
BioPhysical Journal

Using Curriculum Pacing in Learnsphere to Visualize Student Learning Trajectories [Paper] Mar 2019
Sharing and Reusing Data and Analytic Methods with LearnSphere conference

TECHNICAL SKILLS

Languages & Frameworks: Python, R, PostgreSQL, Redis,, C++, Streamlit, Django, Pytorch, Tensorflow, SageMaker, Ollama
AI & Machine Learning: Prompt Engineering, Reinforcement Learning, Quasi Experimental Analysis, RAG, A/B Testing, Causal Inference, Bayesian Inference, Model Interpretability, LangChain, LangGraph, a2a, DeepEval, FAISS, Azure AI, Huggingface
Data Pipelines: Spark, Kafka, Snowflake, Docker, Kubernetes, Jenkins, AWS, Databricks

EDUCATION

Indiana University Bloomington, IN, USA
Masters of Science in Data Science CGPA: 3.93 Aug 2021 – May 2023

Nirma University Ahmedabad, GJ, IN
Bachelors of Science in Computer Science CGPA: 7.8/10 Aug 2014 – June 2018

PROJECTS

Improving Image Captions with Depth Maps [Github]
- Improved image captioning using CNN and **RNN** models and incorporating depth maps, for accurate and informative captions.

Abusive Language Detection in Social Media using Natural Language Processing [Github]
- Developed a multi headed model capable of detecting abusive language and threats like hate speech, obscenity, targeted threats using LSTMs and glove-emoji embeddings

Sliding and Picking objects with Robot Arm using Reinforcement Learning [Github]
- Developed a Q-learning-based control algorithm and implemented a hindsight replay buffer for object manipulation using a robot arm, resulting in a **20-25%** increase in success rates on pushing and sliding tasks.